Should I run AlphaFold?

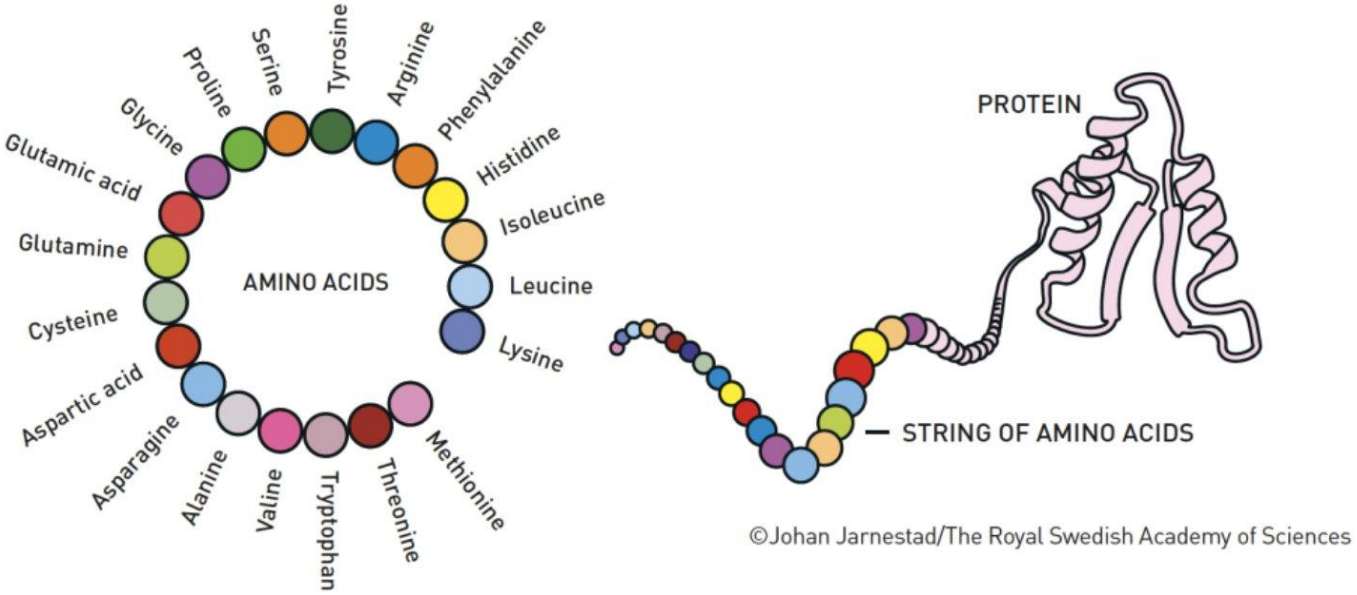
A guide through computational techniques and applicability of structure prediction tools

Christine Schulz
Bio-Structure Hub
SSC, IWR

What is AlphaFold?

Quick Recap

- Prediction of 3D structures of biomolecules based on sequence of amino acids, nucleic acids or SMILES strings



https://www.nobelprize.org/prizes/chemistry/2024/summary/

Nobel Prize in Chemistry 2024



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David Baker

Prize share: 1/2



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Demis Hassabis

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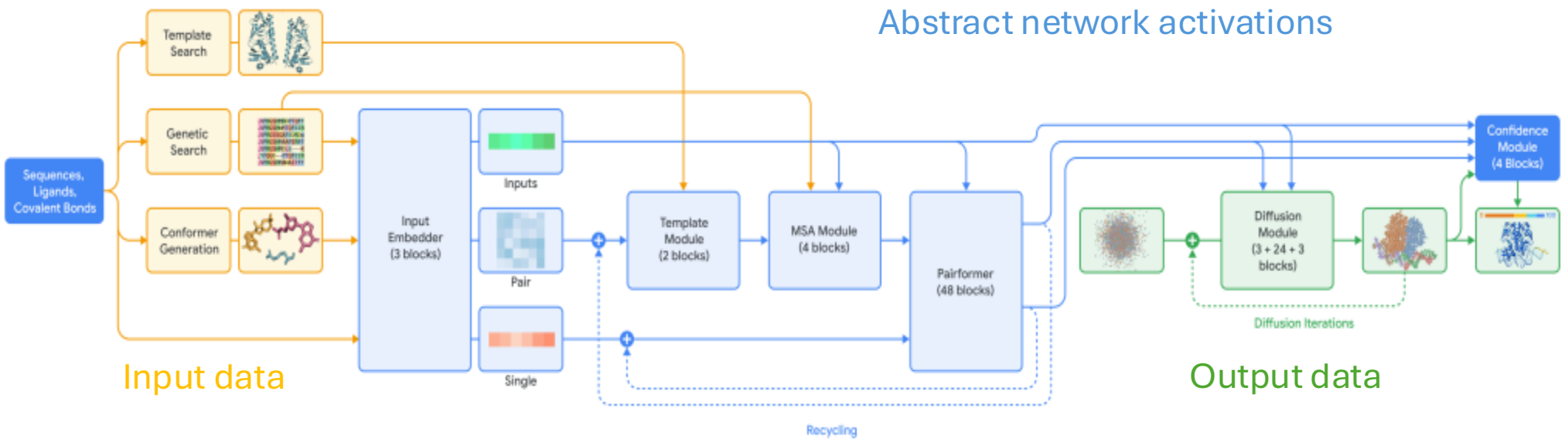
John Jumper

Prize share: 1/4

What is AlphaFold?

Quick Recap

- machine learning model trained on structures in [protein databank]

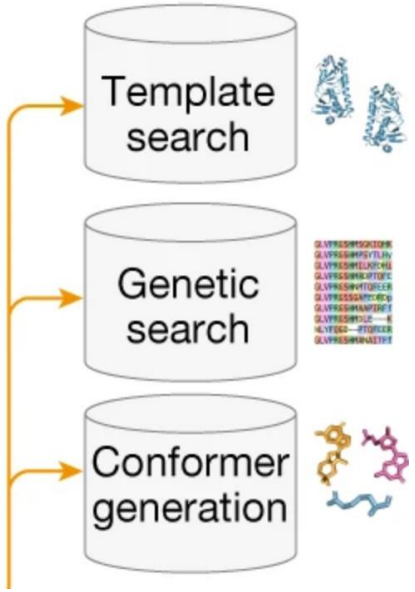


https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/how-does-alphafold-3-work/

What is AlphaFold?

Quick Recap

- machine learning model trained on structures in protein databank



- looking for PDB templates
- Multi-Sequence Alignment (MSA)

Amino acid level, not atom level

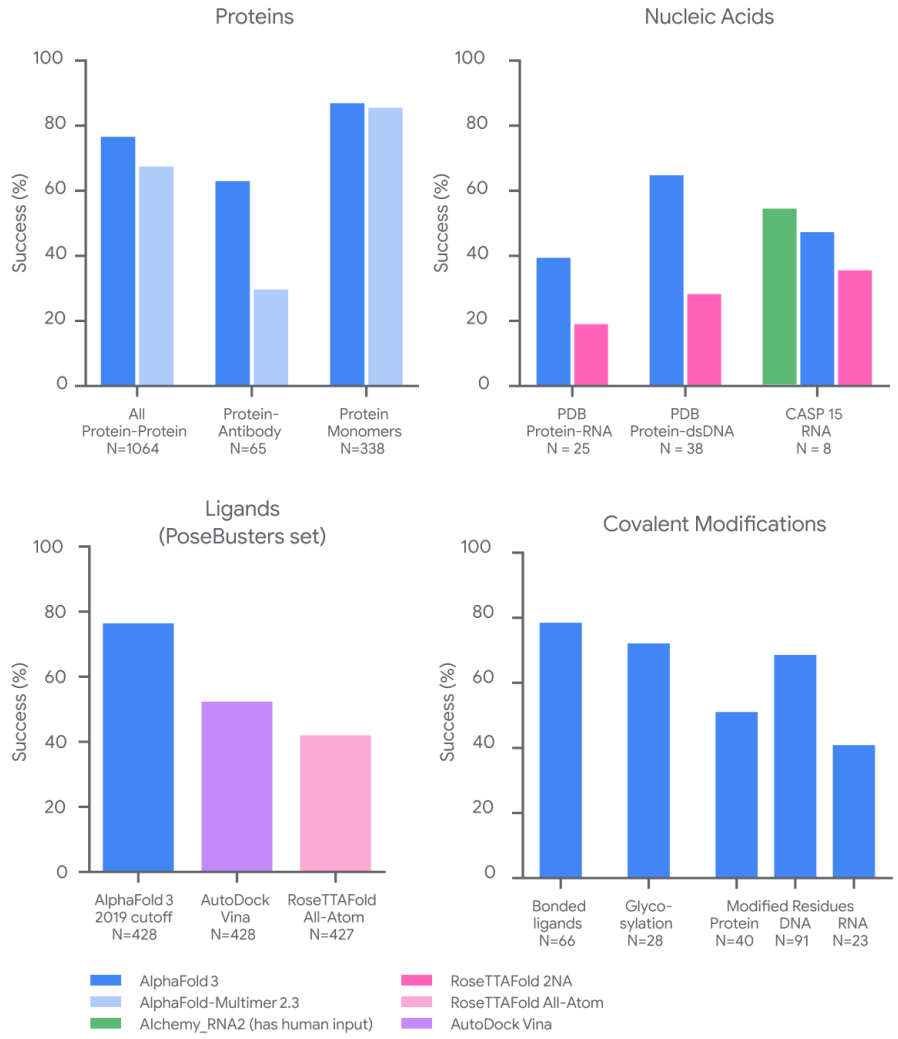
- Small molecule conformations (in AF3)

https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/how-does-alphafold-3-work/

What is AlphaFold good at?

Look at benchmarks

- 3D structures of proteins
  - Monomers, dimers, protein-antibody complexes
- 3D structures of protein-nucleic acid complexes
  - RNA, DNA
- Poses of ligands within protein structures
- Covalent modifications such as bonded ligands, glycosylation

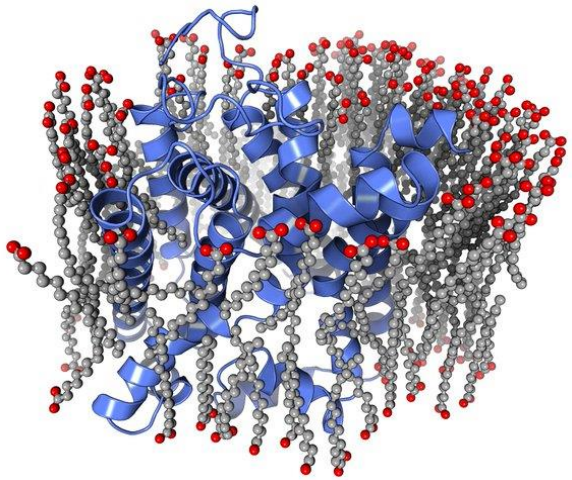


https://www.nature.com/articles/s41586-024-07487-w

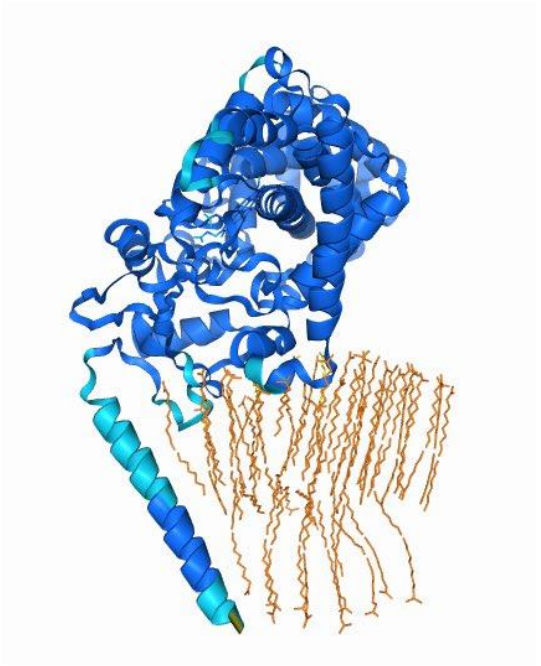
What else can AlphaFold do?

Membrane proteins

- AlphaFold does not know anything about lipids
- Adding a few lipid molecules can lead to a better conformation of trans-membrane regions



https://x.com/fenguita/status/1789177480667959728



https://x.com/caco3cz/status/1788951918867268029

https://www.nature.com/articles/s41586-024-07487-w

What does AlphaFold struggle with?

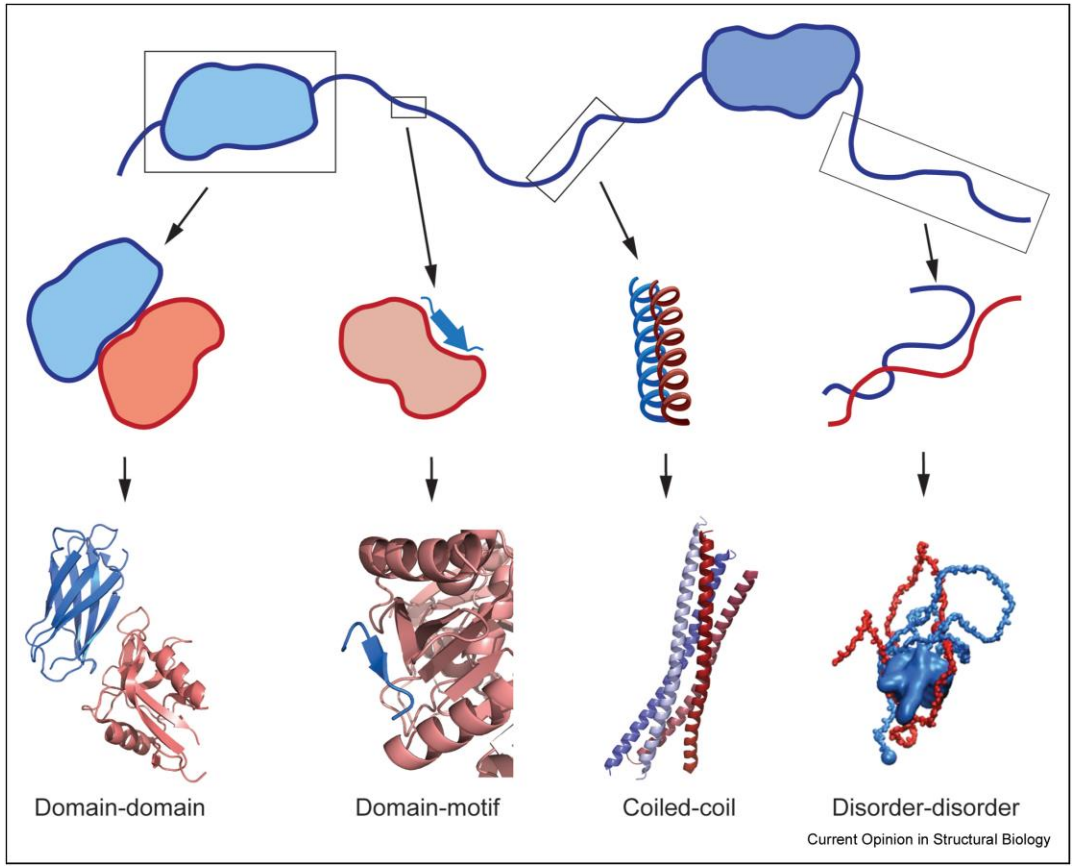
Can AlphaFold predict if two proteins interact?

Problem:

- standard AlphaFold confidence metrics do not reliably separate relevant PPIs from an abundance of false positive predictions (Schmid et al, 2025)
- Protein interfaces including disordered protein regions are underrepresented in training data (Strom et al., 2025)

Workaround:

- Use specific tools to determine PPI probability



https://doi.org/10.1016/j.molcel.2025.01.034
https://doi.org/10.1016/j.sbi.2025.103002

What does AlphaFold struggle with?

Dynamic systems

Problem:

- Protein Data Bank (PDB), and do not capture the dynamic behavior of biomolecular systems in solution.

Result:

- modelled conformational state may not be correct given the specified ligands and other inputs

https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/what-alphafold-3-struggles-with/

What does AlphaFold struggle with?

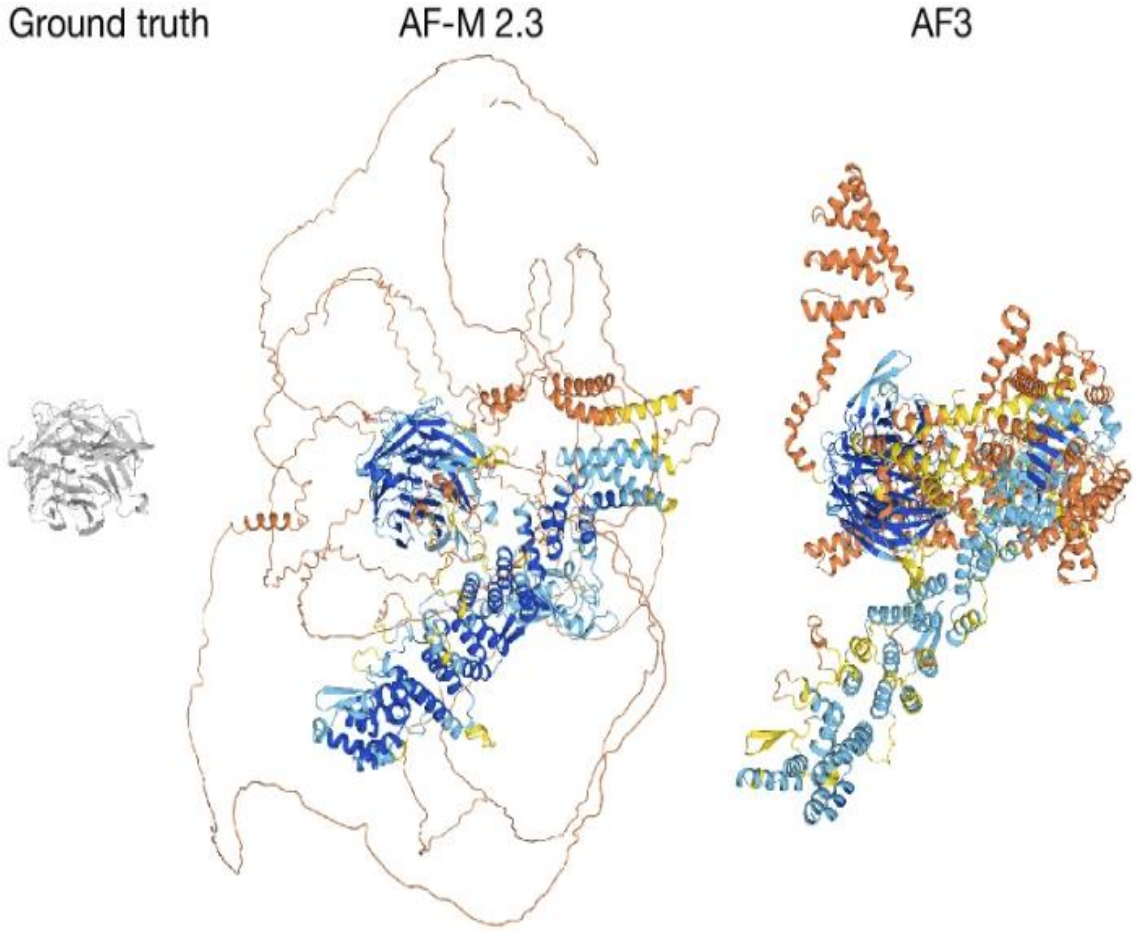
Flexible region

Problem: lack of data on flexible regions in PDB

Result:

- hallucinations in disordered regions
- marked as very low confidence, with pLDDT scores well below 50

- Spaghetti like regions in AlphaFold 2
- Alpha Helix regions in AlphaFold 3



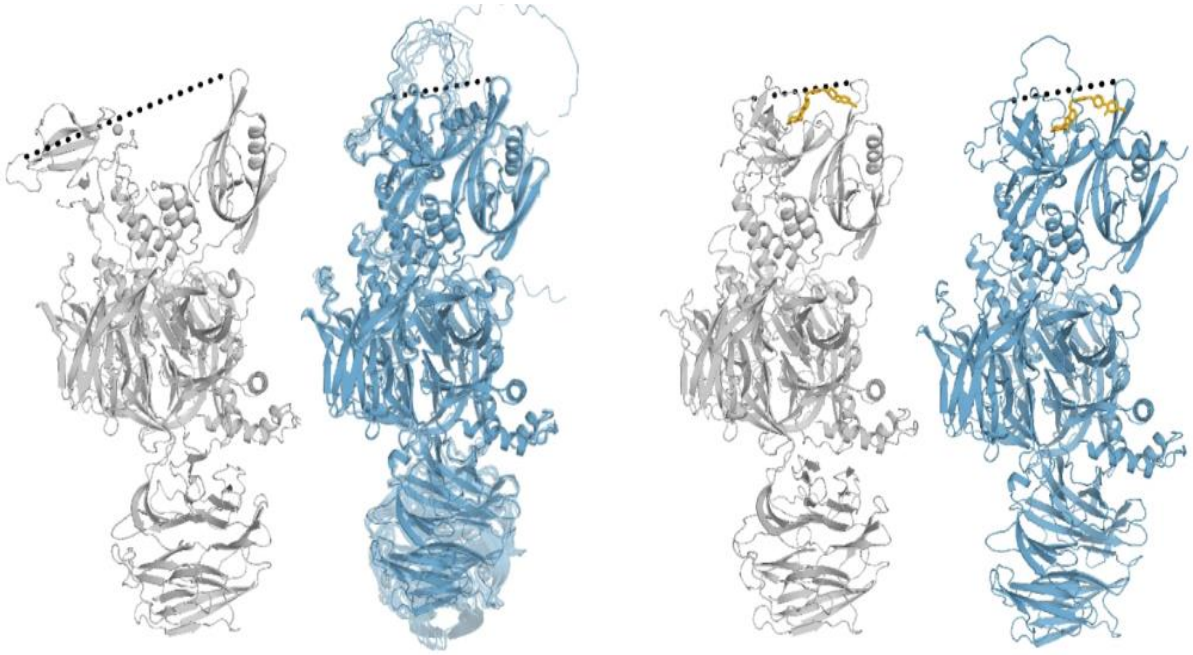
https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/what-alphafold-3-struggles-with/

What does AlphaFold struggle with?

Alternative states

Problem:

- Modelled alternative conformational state may not be correct given the specified ligands and other inputs (e.g. apo vs holo states)



https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/what-alphafold-3-struggles-with/

What does AlphaFold struggle with?

Point Mutations

Problem:

- lack of data on the effect of point mutations on structure

Result:

- not sensitive to point mutations that change a single residue
- "no correlation between AlphaFold output metrics and change of protein stability or [function]" (Pat et al. 2023)

https://www.nature.com/articles/s41586-024-07487-w
https://doi.org/10.1371/journal.pone.0282689

What does AlphaFold struggle with?

Orphaned proteins

Problem: lack of similar sequences (e.g. with antibodies)

Result:

- no sufficiently deep MSAs.

Workaround:

- Sampling: generate a large number of predictions and rank them (costly)
- notably effective for antibody–antigen complexes

https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/what-alphafold-3-struggles-with/

If not AlphaFold, what else..?

Computational Toolbox

Molecular dynamics

- Flexible protein (regions) to generate conformational ensemble https://pubs.acs.org/doi/full/10.1021/acs.chemrev.5b00562
- Influence of point mutations on stability https://doi.org/10.1016/S0006-3495(03)74687-6
- Ligand stability https://doi.org/10.1007/s10822-016-0005-2
- Membrane systems https://doi.org/10.1016/j.sbi.2008.02.003

Meta Dynamics

- Alternative conformational states https://www.nature.com/articles/s42254-020-0153-0

Machine Learning Models

- Protein-protein interactions https://doi.org/10.1016/j.molcel.2025.01.034